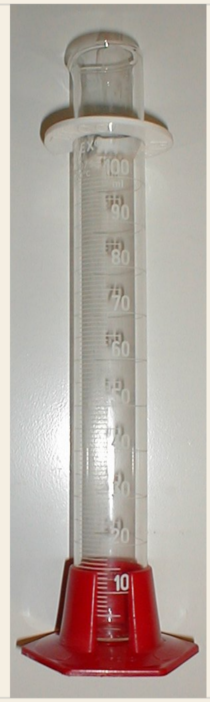
CURIONEST

CHEMISTRY MEASUREMENT

Lab Tools, Density & Percent Error

Visual worksheet for U.S. Grades 8-10



STUDENTS WILL

- identify common measurement tools
- justify equipment choices with visual evidence
- calculate density and percent error
- compare accuracy and precision
- repair a measurement misconception

4 student pages
4-page full key
Editable DOCX

PRINT-READY • US LETTER

density = mass / volume

Photo: Darrien / Haltopub, Public Domain, Wikimedia Commons

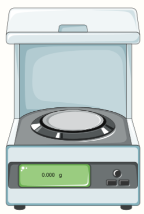
2 Choose the Tool

Name what you see, then use visible features to justify the choice.

Name: ____________________ Date: __________ Class Period: ______

A. Identify each instrument

1

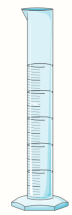


Instrument: ___________________

Quantity/job: _________________

Unit (if measured): __________

2



Instrument: ___________________

Quantity/job: _________________

Unit (if measured): __________

3

Instrument: ___________________

Quantity/job: _________________

Unit (if measured): __________

B. Select and justify

4. You need 18.6 mL of water. Choose a beaker or a graduated cylinder. Explain which visible design features support your choice.

____

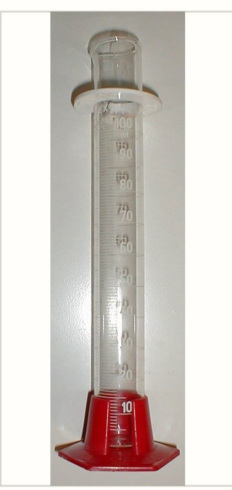
____

5. You need about 125 mL of solution and room to stir. Choose a beaker or a graduated cylinder. Explain.

____

____

C. Read visual evidence



6. Identify two visible features that show this vessel was designed to measure liquid volume.

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____

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TEACHER KEY

3 Calculate with Units

Show the formula, substitute values with units, and round only the final answer.

GUIDED EXAMPLE

A sample has a mass of 36.4 g and a volume of 14.0 mL.
d = m / V = 36.4 g / 14.0 mL = 2.60 g/mL

7. A liquid sample has a mass of 27.0 g and a density of 2.70 g/mL. Calculate its volume.

10.0 mL

8. A liquid has a density of 0.789 g/mL and a volume of 25.0 mL. Calculate its mass.

19.7 g

9. A student measures 2.58 g/mL. The accepted value is 2.70 g/mL. Calculate percent error.

4.44%

10. Accuracy and precision

Accepted density: 2.70 g/mL

Set A	2.70, 2.71, 2.69 g/mL
Set B	2.42, 2.41, 2.42 g/mL

Which set is accurate? Which is precise? Explain using the pattern in the data.

Set A is accurate and precise. Set B is precise but not accurate.